

* ITC *

Information Technology &

(Information Theory & Coding)

- Information techniques the amount of uncertainty / surprise present in any message or statement is known as information.

Terms $P(E) \Rightarrow$ probability of any events

$I(E) \Rightarrow$ Information presents in any event

$$I(E) \propto \frac{1}{P(E)}$$

$$I(E) = \log_b \left(\frac{1}{P(E)} \right)$$

Here $b =$ unit of information

if $b = 2 \Rightarrow$ bits

$b = 10 \Rightarrow$ Hartley

$b = e \Rightarrow$ Nat

Q From a deck of cards, a ^{Red} card is selected randomly. Calculate how much amount of information is collected.

Soln $E =$ Red card

$$P(E) = \frac{26}{52} = \frac{1}{2}$$

then $I(E) = \log_2 \left(\frac{1}{1/2} \right)$

$$I(E) = 1 \text{ bit}$$

ii) How much more info. required to specify exactly the card.

$$E = I = I_1 - I_2$$

$$I_1 = 1 \text{ bit (from previous)}$$

$$\text{for } I_2$$

$$\frac{1}{26} = \frac{1}{26}$$

$$P(E_2) = \frac{1}{26} \quad (26 \text{ Red Cards})$$

$$I_2 = \log_2 \left(\frac{1}{26} \right)$$

$$I_2 = \log_2 (26)$$

$$I = I_2 - I_1$$

$$I = \log_2 (26) - \log_2 (26)$$

$$I = \log_2 \left(\frac{26}{26} \right)$$

$$I = \log_2 (1)$$

Q In a binary system if zero occurs with a probability of $\frac{1}{4}$, 1 occurs with $\frac{3}{4}$ probability. Calculate the amount of info. coded by each bit.

Solⁿ $E_1 = 0$ occurs & $E_2 = 1$ occurs

$$P(E_1) = \frac{1}{4}$$

$$P(E_2) = \frac{3}{4}$$

$$I(E_1) = \log_2(1/1/4)$$

$$I(E_1) = \log_2(4)$$

$$I(E_1) = 2 \text{ bits}$$

$$I(E_2) = \log_2(1/3/4)$$

$$I(E_2) = \log_2(4) - \log_2(3)$$

$$I(E_2) = (2 - \log_2(3))$$

$$I(E_2) = 2 - 1.585$$

$$I(E_2) = 0.415$$

$$+ 2$$

$$0.615$$

Q Calculate the amount information if binary digits occurs with equal likelihood in a binary PCM system.

Solⁿ $E_1 = 50-50\%$ probability

$$P(E_1) = 1/2$$

$$I(E_1) = \log_2(1/1/2)$$

$$I(E_1) = \log_2(1/1/2)$$

$$I(E_1) = \log_2(2)$$

$$I(E_1) = 1 \text{ bit}$$

$$I(E_1) = 1 \text{ bit}$$

Q A random source generates four symbols having the probabilities

$$P(x_1) = 1/4$$

$$P(x_3) = 1/8$$

$$P(x_2) = 1/2$$

$$P(x_4) = 1/8$$

obtain the information contain from each of these messages

$$\text{Sol}^n \quad I(x_1) = \log_2(4) = 2 \text{ bit}$$

$$I(x_2) = \log_2(2) = 1 \text{ bit}$$

$$I(x_3) = \log_2(8) = 3 \text{ bit}$$

$$I(x_4) = \log_2(8) = 3 \text{ bit}$$

Q If there are M equally likely & independent messages, prove that amount of information contain by each message is equal to N bit where $M = 2^N$

$$P(M) = \frac{1}{M}$$

$$\therefore I(M) = \log_2 \left(\frac{1}{P(M)} \right)$$

$$I(M) = \log_2 (M)$$

$$(\because M = 2^N) \quad (\log_{10}(2) = 3.32)$$

$$I(M) = \log_2 (2^N)$$

$$I(M) = N \text{ bit}$$

Ans

Entropy :- Average amount of information.

Q From a deck of cards (52), one card is picked.

A = card is Red

$$P(B) = \frac{1}{52}$$

$$P(C) = \frac{1}{12}$$

is given. calculate information associated with each event

Solⁿ

$$P(A) = \frac{26}{52} = \frac{1}{2}$$

$$\log_2(52) = 3.32 \times \log_{10}(52)$$

$$I(A) = \log_2 \left(\frac{1}{1/2} \right) = 1$$

$$I(B) = \log_2 \left(\frac{1}{1/52} \right) = \log_2(52) = 2 + \log_2(13)$$

$$I(C) = \log_2 \left(\frac{1}{1/12} \right) = \log_2(12) = 2 + \log_2(3)$$

$$= 2.584$$

$$H(X) = - \sum_{i=1}^n p_i \log_2(p_i)$$

$$H(X) = \frac{I(A) + I(B) + I(C)}{3} = \frac{1 + 5.77 + 3.5}{3}$$

$$H(X) = \frac{10.27}{3}$$

$$H(X) = 3.42 \text{ bits}$$

Properties of the entropy :-

(1) An average amount of information

$$(2) 0 \leq H(X) \leq \log N$$

$N = \text{No. of symbols / msgs}$

(3) $H(X)$ is equals to its minimum value
i.e. 0

when $p_i = 1$, then $\log(1) = 0$

$$\therefore H(X) = 0$$

(4) when all the symbols from generated source are equi-probable.

$$H(X)_{\max} = \log N$$

(5) Redundancy

$$\text{red} = \log N - H(X)$$

$$\text{or } 100 - \eta \quad (\text{in } \%)$$

Q In an event of coin tossing if

$$P(H) = 1/2$$

$$P(T) = 1/2$$

find out the entropy

Solⁿ

$$H(X) = \log(N)$$

$$= \sum_{i=1}^2 P_i \log_2(1/P_i)$$

$$= 1/2 \log_2(1/1/2) + 1/2 (\log_2(1/2))$$

$$= \log_2(2) = 1$$

Maximum because of equi-probable.

* Joint Entropy :-

$$P(A, B) = P(A \cap B)$$

if A & B are independent

if A & B are dependent

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

Joint entropy is the entropy associated with the joint probability distribution.

$$H(X, Y) = \sum \sum P(X, Y) \log_2(1/P(X, Y))$$

Q Two biased coins are tossed such that probability of Head (a tail

$$P(H) = 0.4$$

$$P(T) = 0.6$$

Find out the joint Entropy associated with the given probability distribution.

	H	T
H	0.16	0.24
T	0.24	0.36

Solⁿ for dependent Joint probability distribution
 $\{HH, HT, TH, TT\}$

$$H(X, Y) = \sum \sum P(X, Y) \log_2 \left(\frac{1}{P(X, Y)} \right)$$

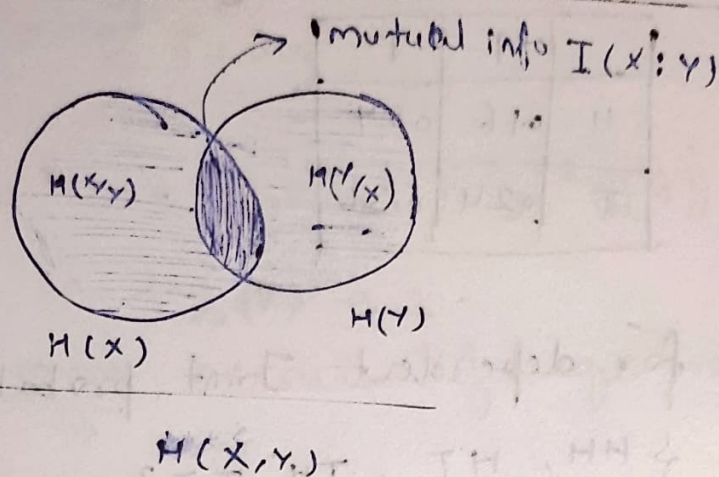
$$H(X, Y) = 0.16 \log_2 \left(\frac{1}{0.16} \right) + 0.24 \log_2 \left(\frac{1}{0.24} \right) + 0.24 \log_2 \left(\frac{1}{0.24} \right) + 0.36 \log_2 \left(\frac{1}{0.36} \right)$$

$$H(X, Y) = 3.32 \left[0.16 + 0.48 + 0.48 + 0.36 \right]$$

$$H(X, Y) =$$

* Conditional Entropy :- Entropy of Random variable of Y if we already know entropy of Random variable X .

$$H(Y/X) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 \left(\frac{1}{P(Y/x)} \right)$$



$$H(X, Y) = H(X/Y) + H(Y/X) + I(X:Y)$$

$$H(X, Y) = H(X) + H(Y/X)$$

$$\text{or } H(Y) + H(X/Y)$$

* Mutual Information ($I(X:Y)$)

If there are two random variables, then the combined information is known as mutual information.

$$I(X:Y) = \sum_{x,y} P(x, y) \log_2 \left[\frac{P(x, y)}{P(x) P(y)} \right]$$

Properties of Mutual information :-

1) Mutual information is symmetric in nature

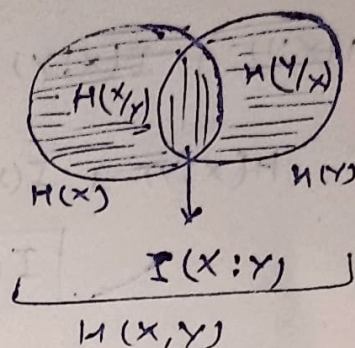
$$I(X; Y) = I(Y; X)$$

2) Mutual information can not be negative

$$\text{i.e. } I(X; Y) \geq 0$$

$$3) I(X; Y) = H(X) - H(X/Y)$$

$$= H(Y) - H(Y/X)$$

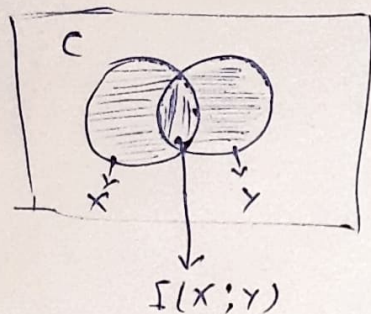


$$4) I(X; Y) = H(X, Y) - H(X/Y) - H(Y/X)$$

(in terms of combine entropy)

* Proof of these equation (HW)

Proof : Prove that Mutual information $I(X; Y) \geq 0$
(always +ive)



$$I(X; Y) = H(X) - H(X/Y)$$

$$I(X; Y) = \sum_{i=1}^m \sum_{j=1}^n P(X_i, Y_j) \log_2 \left(\frac{P(X_i, Y_j)}{P(X_i) P(Y_j)} \right)$$

$$I(X; Y) = - \sum \sum P(X_i, Y_j) \log_2 \left(\frac{P(X_i) P(Y_j)}{P(X_i, Y_j)} \right)$$

use $\frac{1}{n} = \frac{1}{m} = \frac{1}{n}$
($n = \frac{1}{n}$)

$$\log\left(\frac{1}{m}\right) = -\log(m)$$

$\therefore \sum_{k=1}^m P_k \log_2 \left(\frac{P_k}{P_k} \right) \leq 0$
always negative

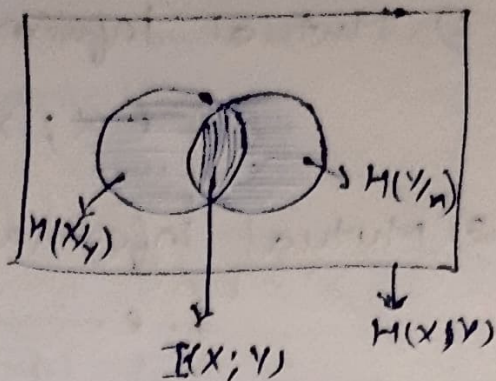
$$I(X; Y) = -[(-)\text{ive}]$$

$$I(X; Y) = (+)\text{ive} \Rightarrow \boxed{I(X; Y) \geq 0}$$

Q Prove that $I(X; Y) = H(X) + H(Y) - H(X, Y)$

Solⁿ $H(X, Y) = H(X/Y) + H(Y) - I$

Q $I(X; Y) = H(X) - H(X/Y)$
 add both



$$H(X, Y) + I(X; Y) = H(X/Y) + H(Y) + H(X) - H(X/Y)$$

$$H(X, Y) + I(X; Y) = H(X) + H(Y)$$

$$\boxed{I(X; Y) = H(X) + H(Y) - H(X, Y)} \quad \text{H.P.}$$

* Channels :- Transmission medium by which information (signals) pass from source to destination.

Types of channels -

- 1) Lossless channel
- 2) Deterministic channel
- 3) Noiseless channel
- 4) Binary Symmetric
- 5) channel (BSC)

DMS (Discrete Memoryless source)

The Discrete memoryless channel (DMC) is a connecting medium between the transmitter & receiver. A DMC is represented by channel matrices

$$P \left[\begin{matrix} Y_m \\ X_n \end{matrix} \right]$$

Here

Y_m = outputs

X_n = number of inputs

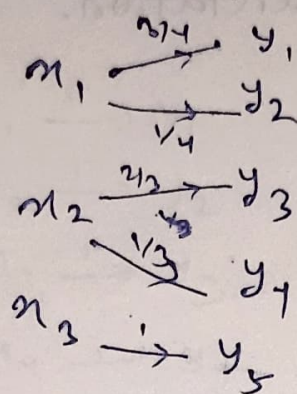
$$P \left[\begin{matrix} Y_m \\ X_n \end{matrix} \right]$$

$$= \begin{matrix} & \begin{matrix} Y_1 & Y_2 & Y_3 & Y_4 & \dots & Y_m \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ \vdots \\ X_n \end{matrix} & \begin{bmatrix} P(Y_1/X_1) & P(Y_2/X_1) & P(Y_3/X_1) & P(Y_4/X_1) & \dots & P(Y_m/X_1) \\ P(Y_1/X_2) & P(Y_2/X_2) & P(Y_3/X_2) & P(Y_4/X_2) & \dots & P(Y_m/X_2) \\ P(Y_1/X_3) & P(Y_2/X_3) & P(Y_3/X_3) & P(Y_4/X_3) & \dots & P(Y_m/X_3) \\ P(Y_1/X_4) & P(Y_2/X_4) & P(Y_3/X_4) & P(Y_4/X_4) & \dots & P(Y_m/X_4) \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ P(Y_1/X_n) & P(Y_2/X_n) & P(Y_3/X_n) & \dots & \dots & P(Y_m/X_n) \end{bmatrix} \end{matrix}$$

1) Lossless channel:- let us assume

There are three routes n_1, n_2, n_3 applied to the DMC. As the result, we are getting 5 outputs. y_1, y_2, y_3, y_4, y_5

The channel matrix is given by following graphical representation of the channel



channel matrix

$$P[y_j/x_i] = \begin{matrix} & y_1 & y_2 & y_3 & y_4 & y_5 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 3/4 & 1/4 & 0 & 0 & 0 \\ 0 & 0 & 2/3 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

* A channel which has only 1 non-zero element in each column. These known as lossless channel.

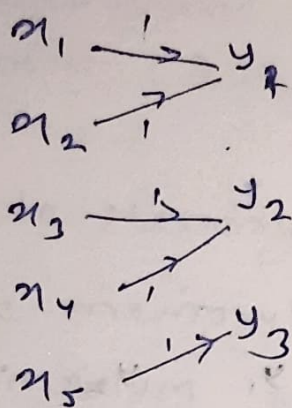
2) Deterministic Channel

A channel matrix which is having only one ~~known~~^{non} zero element in each row, which must be equal to unity, is known as deterministic channel.

Let us assume there are 5 inputs

x_1, x_2, x_3, x_4, x_5 & 3 outputs: y_1, y_2, y_3

then graphical representation,



Then channel matrix

$$\begin{matrix} & y_1 & y_2 & y_3 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

3) Noiseless channel:-

A channel matrix that has characteristics of the noiseless both deterministic & lossless channel. That means each row & column will have only one known zero element that is 1.

Let us assume inputs are x_1, x_2, x_3, x_4 & outputs are y_1, y_2, y_3, y_4

matrix

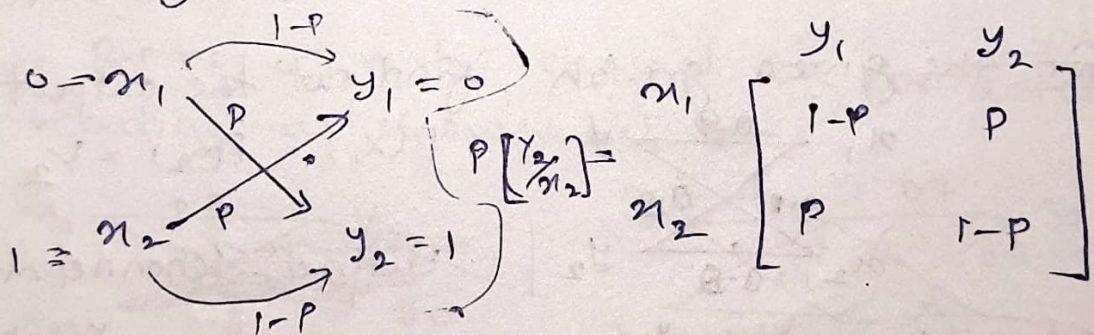
$$P[Y_j/X_i] = \begin{matrix} & y_1 & y_2 & y_3 & y_4 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$x_1 \xrightarrow{1} y_1$
 $x_2 \xrightarrow{1} y_2$
 $x_3 \xrightarrow{1} y_3$
 $x_4 \xrightarrow{1} y_4$

Note Number of inputs & number of outputs are equal.

4) Binary Symmetric Channel:-

A channel is symmetric if the probability of receiving 1, if binary zero (0) is sent is equal to the probability of receiving zero if binary one (1) is sent.



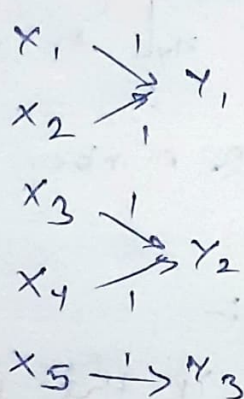
Q1 for the given following channel matrices

$$\begin{matrix} x_1 & x_2 & x_3 \end{matrix} \begin{matrix} y_1 & y_2 & y_3 & y_4 & y_5 \\ \begin{bmatrix} 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 2/5 & 3/5 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Find out the type of channel.

Soln In each column, the channel matrices contains only one unique (non zero) element. So it is lossless channel.

Q2 for the given channel matrices find out the types of the channel.

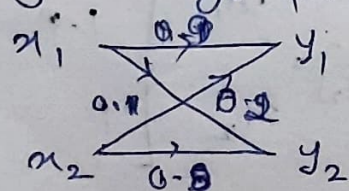


Soln

$$\Rightarrow \begin{matrix} y_1 & y_2 & y_3 \\ \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Each row contains only one unique non-zero element thus, it is Deterministic channel matrix.

Q3 for the given graph, find out the output if



$$P(x_1) = P(x_2) = 1/2$$

output = (channel matrix of)
x(input)

$$\Rightarrow \begin{matrix} y_1 & y_2 \\ \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \end{matrix}$$

$$\text{output} = \text{i/p} \times \text{channel matrices}$$

$$\Rightarrow \text{output} = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

$$\Rightarrow \text{output} = [0.45 + 0.10 \quad 0.05 + 0.40]$$

$$\Rightarrow \text{output} = [0.55 \quad 0.45]$$

$$y_1 = 0.55$$

$$\& y_2 = 0.45 \quad \text{Answer}$$

Q If input variable $x_1 = 0.5$, $x_2 = 0.5$ & the channel matrices is given as

$$\begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

Find out the output.

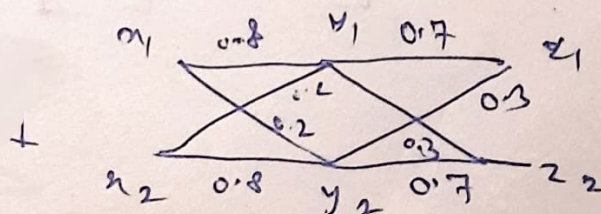
$$\text{Sol}^n \quad \text{output} = \text{i/p} \times \text{channel matrices}$$

$$\text{output} = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix} \times \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}$$

$$\text{output} = [0.5 \quad 0.5]$$

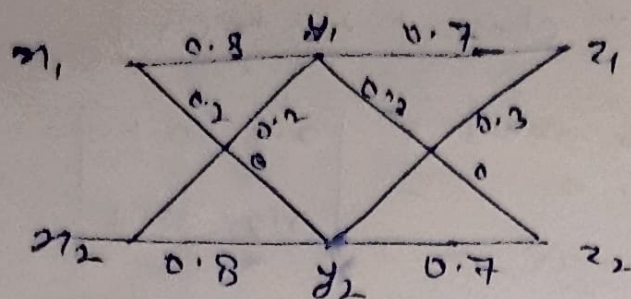
$$y_1 = 0.5 \& y_2 = 0.5 \quad \text{Answer}$$

Q two Binary symmetric channel are connected in cascading as shown below -



find out combined channel matrices.

Soln



$$P(Y/X) = \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} P(y_{x_1}) & P(y_{x_2}) \\ P(y_{x_1}) & P(y_{x_2}) \end{bmatrix} \end{matrix}$$

So first of all find y_1 & y_2

$$x \text{ & } y: \begin{matrix} & y_1 & y_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

$$P(Z/Y) = \begin{matrix} & z_1 & z_2 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} P(z_{y_1}) & P(z_{y_2}) \\ P(z_{y_1}) & P(z_{y_2}) \end{bmatrix} \end{matrix}$$

output = input $\times$ matrices.

$$y \text{ & } z = \begin{matrix} & z_1 & z_2 \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \end{matrix}$$

& x to z

$$P(Z/X) = \begin{matrix} & z_1 & z_2 \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \end{matrix}$$

$$P(Z/X) = \begin{bmatrix} 0.56 + 0.06 & 0.24 + 0.14 \\ 0.14 + 0.24 & 0.06 + 0.56 \end{bmatrix}$$

$$P(Z/X) = \begin{bmatrix} 0.62 & 0.38 \\ 0.38 & 0.62 \end{bmatrix}$$

(ii) find out the output: $P(z_1)$ & $P(z_2)$ if

$$P(x_1) = 0.6$$

$$P(x_2) = 0.4$$

Solⁿ output = input $\times$ channel matrices.

$$P(z) = P(x) \times$$

$$P(z) = P(x) \times \begin{bmatrix} 0.62 & 0.38 \\ 0.38 & 0.62 \end{bmatrix}$$

$$P(z) = [0.6 \ 0.4] \begin{bmatrix} 0.62 & 0.38 \\ 0.38 & 0.62 \end{bmatrix}$$

$$P(z) = \begin{bmatrix} 0.524 & 0.476 \\ \downarrow & \downarrow \\ z_1 & z_2 \end{bmatrix}$$

* Channel Capacity:- $C = B \log_2(1 + S/N)$

S/N = Signal to Noise ratio
 B = Bandwidth

Q For a typical telephone network S/N ratio is given as 30dB. Is the bandwidth of 3KHz, find out the max data rate (bps).

Sol: Bandwidth = 3KHz

$S/N = 30\text{dB} = 10^3$

$C = B \log_2(1 + S/N)$

$C = 3 \times 10^3 \log_2(1 + 10^3)$

$C = 3 \times 10^3 \log_2(1001)$

$C = 3 \times 10^3 \times \frac{3.004}{0.3010}$

$C = 0.0293 \text{ Mbps}$

$C \approx 0.03 \text{ Mbps}$

$10\text{dB} = 10$

$20\text{dB} = 10^2$

$30\text{dB} = 10^3$

Q For a satellite TV channel signal to Noise Ratio is given as $S/N = 20\text{dB}$, Bandwidth of 10MHz. Rate?

Sol: Given $S/N = 20\text{dB} = 10^2$

$B = 10\text{MHz}$

$C = B \log_2(S/N + 1)$

$C = 10 \times 10^6 \log_2(10^2 + 1)$

$C = 10 \times 10^6 \times \frac{\log(101)}{\log(2)}$

$C = 10 \times 10^6 \times 6.6585$

$C \approx 66.58 \text{ Mbps}$

Q A gaussian channel has the bandwidth of 1MHz, calculate the channel capacity if the signal power to the Noise density ratio $= 10^5$.

Solⁿ $B = 1 \text{ MHz}$
 $S/N = 10^5$ (already converted)

$$C = B \log_2 (1 + S/N)$$

$$C = 1 \times 10^6 \log_2 (1 + 10^5)$$

$$C = \frac{1 \times 10^6 \times 5.0000043}{0.3010}$$

$$C = 10^6 \times 16.6113$$

$$\boxed{C = 16.61 \text{ Mbps}}$$

Int^Q Find out the channel capacity with $B \rightarrow \infty$.

Solⁿ Let us assume noise power $N = \eta B$
where $\eta/2$ = noise power spectral density

$$\text{So } C = B \log_2 \left(1 + \frac{S}{\eta B} \right)$$

$$C = B \frac{\eta}{S} \times \frac{S}{\eta} \log_2 \left(1 + \frac{S}{\eta B} \right)$$

$$\therefore \eta B = N$$

$$C = B \frac{S}{N} \log_2 \left(1 + \frac{S}{\eta B} \right)$$

$$\therefore \lim_{n \rightarrow \infty} \log_2 (1 + \frac{S}{nB})^{nB} = \log_2 e$$

$$\Rightarrow C = B \cdot \frac{S}{n} \log_2 (1 + \frac{S}{nB}) \cdot \frac{nB}{S}$$

$$\text{let } \frac{nB}{S} = n$$

$$\Rightarrow C = \frac{S}{n} \lim_{B \rightarrow \infty} \log_2 (1 + \frac{S}{nB}) \cdot \frac{nB}{S}$$

$$\Rightarrow \boxed{C = \frac{S}{n} \log_2 e}$$

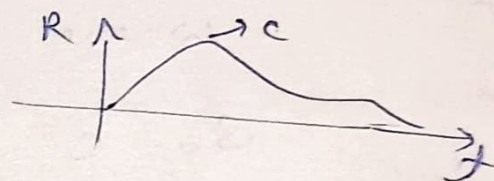
$$\Rightarrow \boxed{C = 1.44 \frac{S}{n}}$$

* Shannon Hartley Channel Capacity Theorem:-

This theorem relates the channel capacity with the bandwidth of the channel & SNR (Signal to Noise Ratio) of the channel.

Capacity of the channel $\rightarrow$ It is the maximum information rate which is travelling from transmitter to the receiver of the channel. It is denoted by 'C'.

$R \Rightarrow$ information Rate
 $C \Rightarrow$ max of R



There are two conditions

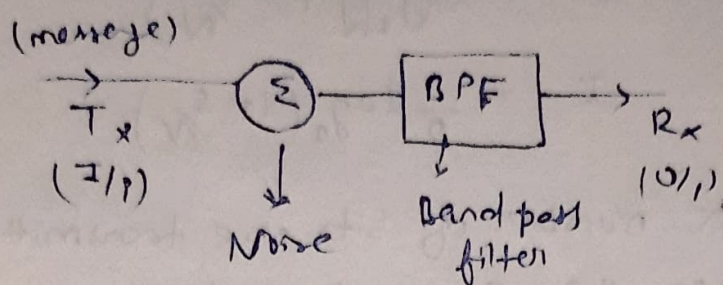
1. $R < C$
2. $R > C$

for $R < C$, in this case, transmission is possible. Errors can be removed by using some algorithms.

for $R > C$, in this case, error free transmission is not possible.

To get the error free transmission, some error correcting & detecting codes are used, which will increase the length of the message as

well as it will increase the complexity of the circuit



So at the output of the band Pass filter we will get signal + Noise.

Let us assume V_s is rms value of the signal voltage.

$\rightarrow V_N$ is rms value of the Noise voltage.

Let us assume Antenna resistance

$$R = 1 \Omega$$

Then $S = \text{signal power} = \frac{V_s^2}{R}$

$$S = V_s^2$$

Similarly $N = \text{Noise power} = \frac{V_N^2}{R}$

$$N = V_N^2$$

from these equation $V_s = \sqrt{S}$ & $V_N = \sqrt{N}$

Maximum number of Levels by which information is transmitted is given as $m = \frac{\sqrt{S+N}}{\sqrt{N}}$

$$m = \sqrt{1 + S/N}$$

So information I is given as

$$I = \log_2 M$$

$$I = \log_2 \sqrt{1 + \frac{S}{N}}$$

$$I = \frac{1}{2} \log_2 (1 + \frac{S}{N})$$

If K number of bits are transmitting per second then the channel capacity C is given as

$$C = \frac{K}{2} \log_2 (1 + \frac{S}{N}) \quad (C = KI)$$

If the bandwidth of the channel is B then maximum information Rate = $2B$

$$K = 2B$$

$$C = \frac{2B}{2} \log_2 (1 + \frac{S}{N})$$

$$\boxed{C = B \log_2 (1 + \frac{S}{N})}$$

$$* \left\{ C \propto B \quad \& \quad C \propto \frac{S}{N} \right\} *$$